

Econometrics I Mock Exam

Matías Cabello – November 24, 2025

Exam duration: 60 minutes. Total points: 60 (one per minute). Consider the points assigned to each question to manage your time. Read carefully and try to answer precisely. You may use formulas or graphs if you wish. Good luck!

A. Mathematical Questions

(A1) $\mathbb{E}[uu'|X] = \sigma^2 I$ is assumed for obtaining the variances of the OLS estimator.

- i. (3 points) What is the meaning of $\mathbb{E}[uu'|X] = \sigma^2 I$?
- ii. (8 points) Provided that $\hat{\beta} = \beta + (X'X)^{-1}X'u$ is unbiased under $y = X\beta + u$ and $\mathbb{E}(u|X) = 0$, how do you arrive at $\mathbb{E}[(\hat{\beta} - \mathbb{E}[\hat{\beta}])(\hat{\beta} - \mathbb{E}[\hat{\beta}])'|X] = \sigma^2(X'X)^{-1}$?
Hint: start from $\hat{\beta} = \beta + (X'X)^{-1}X'u$, assume unbiasedness, and plug $\mathbb{E}[uu'|X] = \sigma^2 I$ when needed.

(A2) You estimate the model $y = X\hat{\beta} + \hat{u}$ through the estimator $\hat{\beta} = (X'X)^{-1}X'y$. However, the data generating process is $y = X\beta + A\gamma + \varepsilon$, with $\mathbb{E}(\varepsilon|X) = 0$.

- i. (1 point) What is the technical name of this problem?
- ii. (6 points) Show that your estimate will be unbiased when $\mathbb{E}(X'A|X) = 0$.

B. True, ambiguous, or false? Argue *briefly* whether each of the following statements is true, false, or uncertain based on technical arguments, graphs, and/or formulas.

- (B1) (3 points) In the classical regression model, the inclusion of irrelevant variables does not produce any bias but it reduces efficiency.
- (B2) (3 points) You are estimating an univariate OLS regression and found an observation with a residual that is 10 times higher than all others. Dropping it from the regression will certainly change the slope of your estimate.
- (B3) (3 points) Normally distributed errors are a precondition for the t-statistics to make sense, either with small or infinitely large sample sizes.
- (B4) (3 points) In an two-stages least squared instrumental variable estimation, we first regress the treatment of interest against all other remaining regressors, then regress the dependent variable on the resulting residuals.

C. Empirical Case Study. Table 1 shows a series of estimates of the effect of maternal tobacco consumption on newborn birth weight. Answer the following questions based on those results.

- (C1) (4 points) What are $\hat{\beta}$ and $\hat{\sigma}^2(X'X)^{-1}$ according to column (2).
- (C2) (5 points) What is the expected weight of a female newborn if the mother is non-smoker? What is the expected 95% confidence interval? Do not compute the value,

just state it arithmetically.

- (C3) It is suspected that a lower number of cigarettes is associated with a higher dispersion of birth weights.
- i. (1 point) What result in column (1) may be compromised?
 - ii. (4 points) What two possible remedies would you suggest?
 - iii. (4 points) How could you verify this concern using a statistical test?
- (C4) (4 points) The standard error of the cigarettes coefficient is lower in (2) than in (1), but higher in (3) than in (2). How do you explain this phenomenon?
- (C5) (4 points) The data of table 1 was produced in a nonexperimental setting. Why should this be a concern for estimating the effect of smoking on birth weights?
- (C6) (4 points) Which of the three estimates seems more appropriate to you for quantifying the impact of a mother consuming one additional cigarette per day on a baby's birth weight? Justify.

Table 1: OLS regressions

Dependent variable: Birth weight (grams)			
	(1)	(2)	(3)
Constant	3421.7*** (14.15)	3375.6*** (19.96)	2243.4*** (397.2)
Cigarettes per day	-11.48*** (3.245)	-11.40*** (3.236)	-10.30** (3.327)
Male baby (dummy)		89.20** (27.33)	85.09** (27.95)
Mother's age, years			66.51* (27.16)
(Mother's age) ²			-1.050* (0.454)
No. of prenatal visits			11.50** (3.753)
Mother's educ, years			-1.784 (7.167)
Observations	1722	1722	1644
R^2	0.007	0.013	0.023
Adjusted R^2	0.007	0.012	0.019

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$